

# From Face Perception to Social Categorization: A Multidimensional Scaling Approach to Ethnic Outgroup Homogeneity in Childhood

Luca Cussigh, Luciana Carraro, Luigi Castelli



UNIVERSITÀ  
DI PADOVA



# The Outgroup Homogeneity Effect (OHE; Quattrone & Jones, 1981)

Outgroup members appear more homogeneous to each other than ingroup members

- Supported in adult populations

# The Outgroup Homogeneity Effect

## Early Emergence of Categorization vs. Childhood OHE Gap

- **Categorization in childhood:** Strong evidence for ingroup preferences and visual specialization for ingroup faces (Kelly et al., 2009; Quinn et al., 2016).
- **OHE in childhood:** No consolidated evidence for the Outgroup Homogeneity Effect (OHE). Why?

## A few structural obstacles

1. Difficult not to employ explicit tasks (creating top-down categories)
2. Most of the literature employs a dyadic paradigm (ingroup-outgroup)
3. If visual paradigms are used, the stimuli selection is subject to biases

# 1. Spontaneous bottom-up categorization

## Multidimensional Scaling (MDS)

- Face-Space Theory (Valentine et al., 2016): Faces exist in a continuous multidimensional perceptual space
- **Goal:** Visually map whether faces from different ethnic groups are spontaneously perceived as more similar to one another (OHE), and how this changes across age groups.

## 2. Multi-group context

- 4-group context: Italian majority-ingroup + 3 distinct ethnic minority-outgroups (African Descent, East Asian, Andean)
- 20 faces per gender group (5 per ethnic group)

### 3. The AI generated-stimuli

- Morphologies controlled using an emotional filter and FaceNet embeddings to ensure similar perceptual distinctiveness across groups

| African Descent                                                                    | East-Asian                                                                          | Italian                                                                              | Andean                                                                               |
|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|
|  |  |  |  |

# The Face Similarity Task

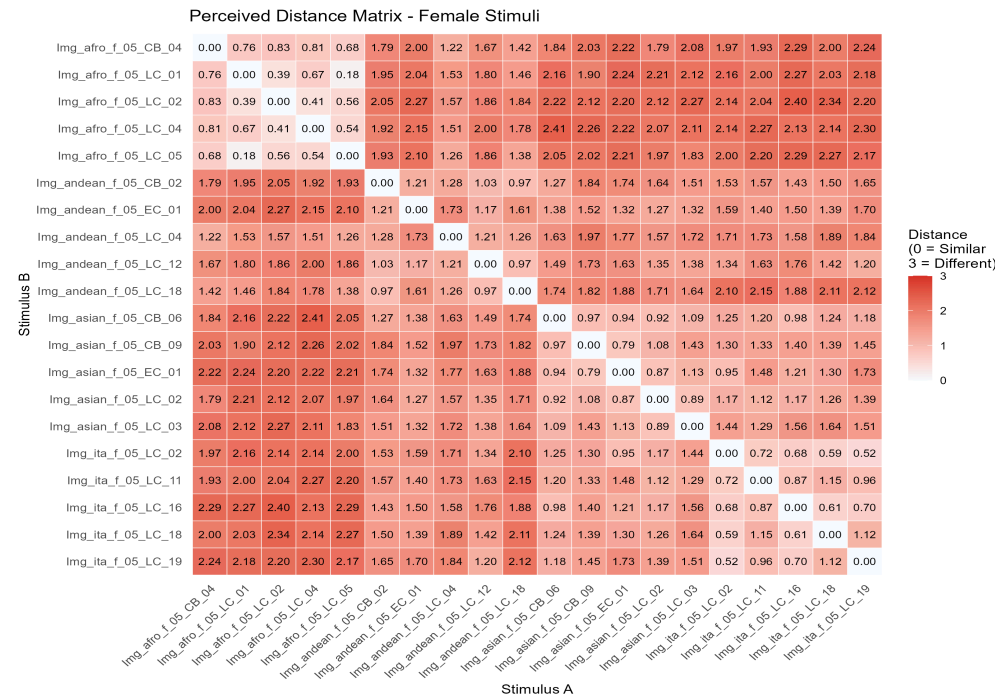
| →                                                                                 | →                                                                                   | →                                                                                                                                        |
|-----------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------|
|  |  | <p>Quanto si somigliano?</p> <div><div>Molto Diversi</div><div>Un po' Diversi</div><div>Un po' Simili</div><div>Molto Simili</div></div> |
| 800ms                                                                             | 1000ms                                                                              | //                                                                                                                                       |



# The Face Similarity Task

- Incomplete block design
- 66 random dyads presented sequentially (3 faces per group)

## Moderated (S1)



# Pre-registered studies: Method & Experimental Design

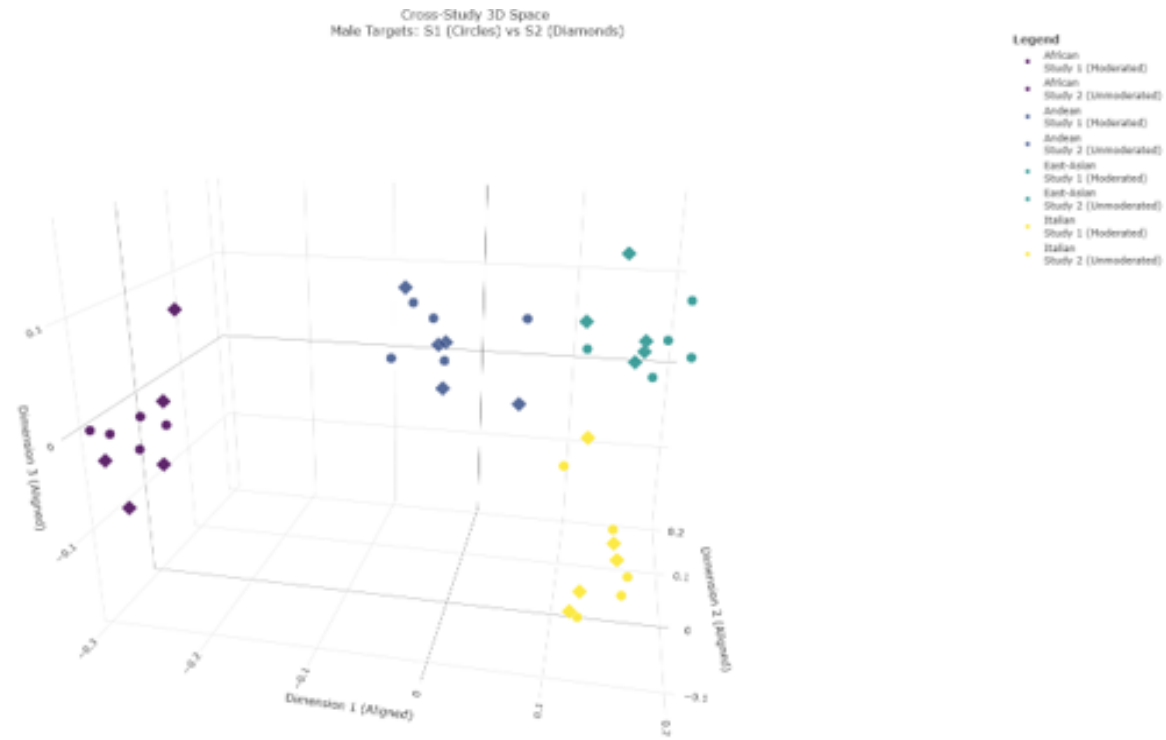
## Sample:

- Study 1 (moderated, controlled school setting):  $N=249$ ,  $M=5.17$  years,  $SD=1.23$ , 134 boys (53.8%) and 115 girls (46.2%)
- Study 2 (unmoderated, online setting):  $N=101$ ,  $M = 6.06$ ,  $SD=1.43$ , 52 girls (51.5%) and 49 boys (48.5%)
- Age range: 3 to 8 years

# MDS

3D MDS visualization

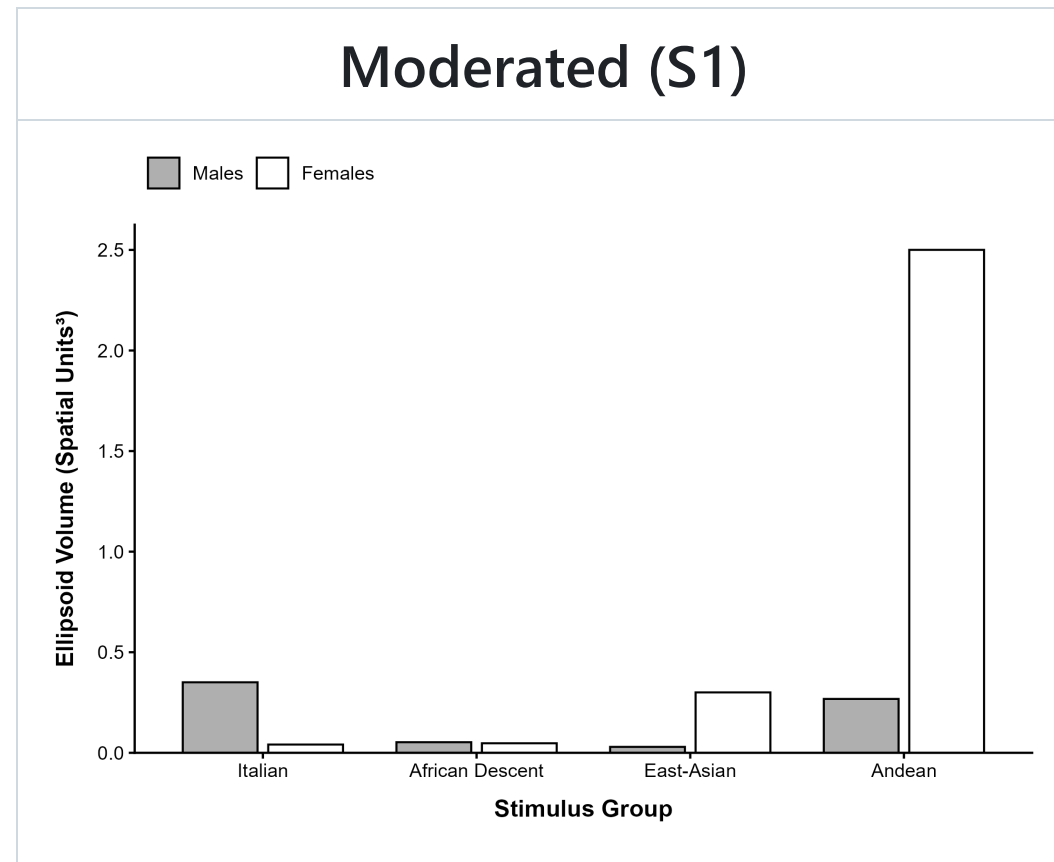
# Moderated (S1)



- Children spontaneously visually categorize stimuli in ethnic groups

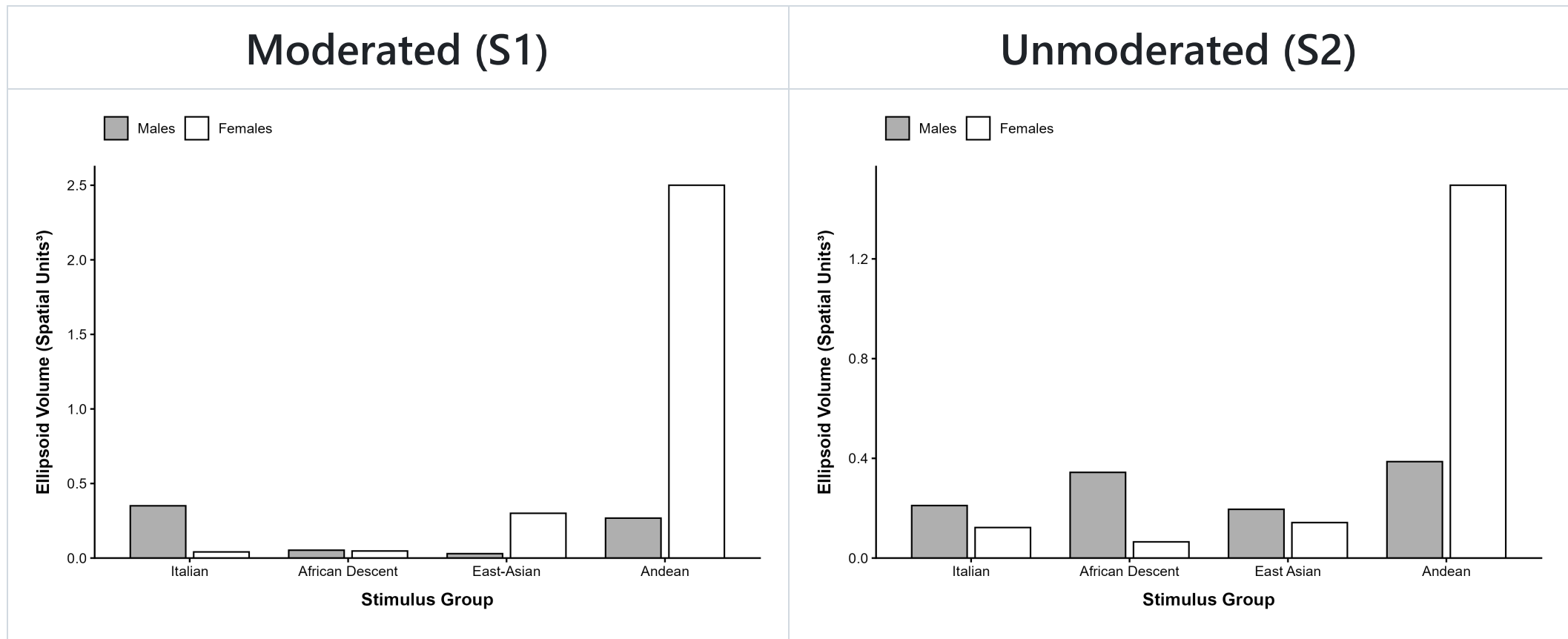
# Outgroup Homogeneity Effect

- Ellipsoid's volume:



# Outgroup Homogeneity Effect

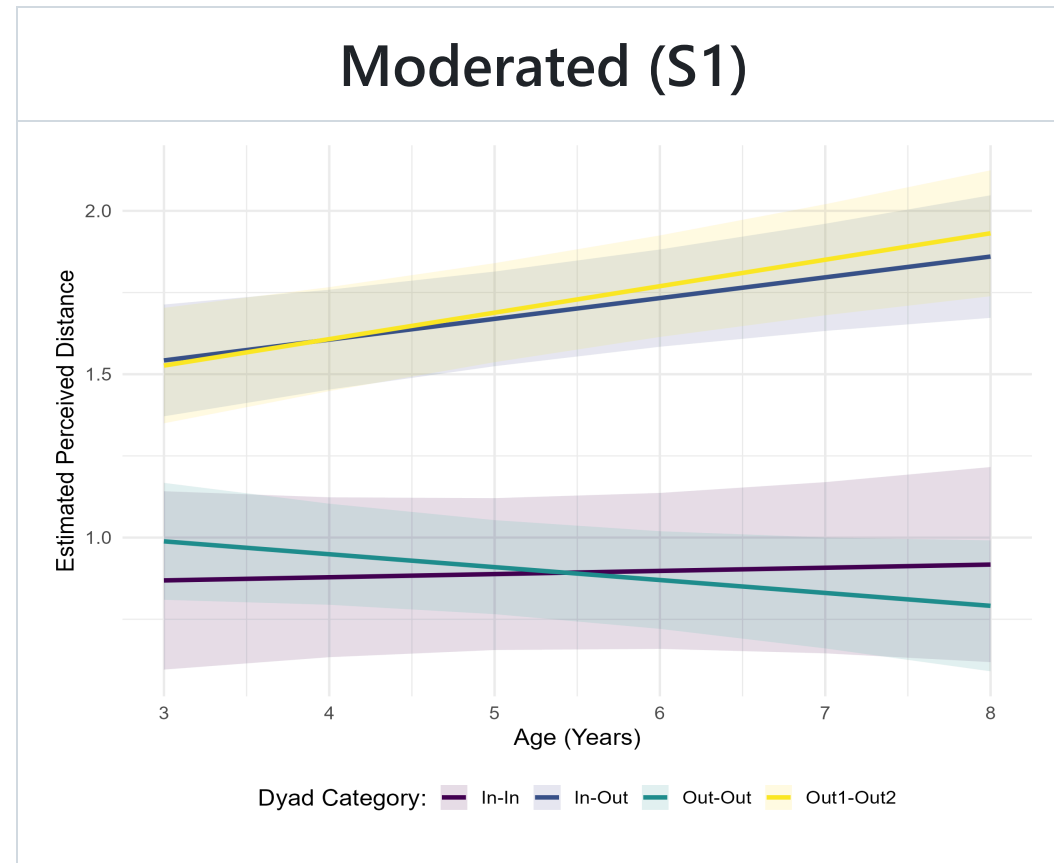
- Ellipsoid's volume:



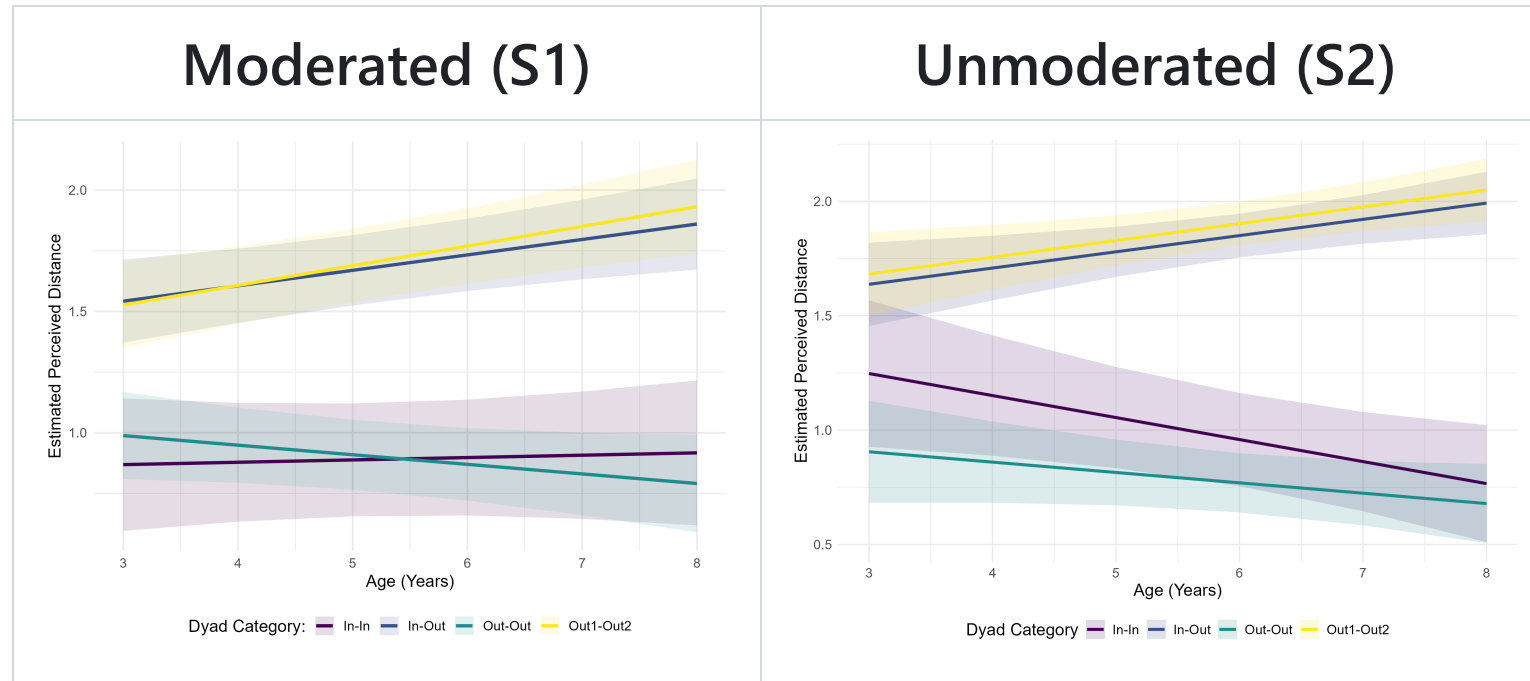
# LMM - Developmental Trajectories



# LMM - Developmental Trajectories



# LMM - Developmental Trajectories



| Dyad type | <i>b</i> (S1) | 95% CI (S1) | <i>b</i> (S2) | 95% CI (S2)  |
|-----------|---------------|-------------|---------------|--------------|
| In-Out    | 0.06          | 0.02, 0.11  | 0.07          | 0.02, 0.12   |
| Out1-Out2 | 0.08          | 0.04, 0.12  | 0.07          | 0.02, 0.12   |
| In-In     | 0.01          | -0.06, 0.08 | -0.10         | -0.18, -0.02 |
| Out-Out   | -0.04         | -0.09, 0.01 | -0.05         | -0.10, 0.01  |

- Intergroup similarity judgments increased with age

# Conclusion

- Children spontaneously visually categorize stimuli in ethnic groups
- Intergroup similarity judgments increased with age

Questions? 🙌

PRESENTATION AVAILABLE AT

